Activity 9

1. Consider the path between point A and point B shown in Fig. 3. The length of the path is given by:

$$L = \int_A^B ds$$

We are interested in finding the right function $y(x)$ that connects A and B and minimizes the length of the path L between A and B. We know the fight answer is that this is a line, but we're going to try to prove it.

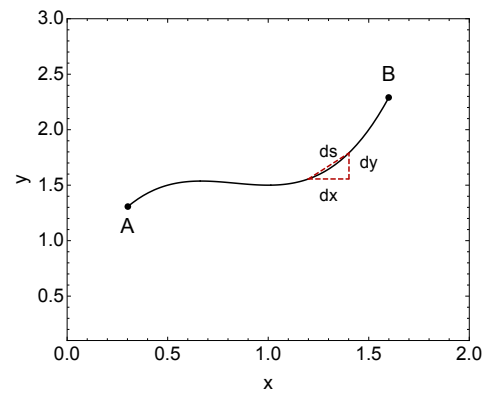

Figure 1: The path between Point A and Point B.

(a) Write ds in terms of dx and dy.

$ds = \sqrt{dx^2 + dy^2}$

$ds^2 = d\vec{r} \cdot d\vec{r}$

$d\vec{r} = dx\,\hat{x} + dy\,\hat{y}$

(b) Rewrite L as an integral from x_A to x_B, where the integration is in terms of dx.

$$L = \int_A^B ds = \int_A^B \sqrt{dx^2 + dy^2} = \int_A^B \sqrt{dx^2}\sqrt{1 + \frac{dy^2}{dx^2}} = \boxed{\int_{x_A}^{x_B} dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}}$$

(c) Rewrite L in the form of

$$L = \int_{x_A}^{x_B} f[y(x), y'(x), x]dx$$

$\frac{dy}{dx} = y'(x)$

$$\boxed{L = \int_{x_A}^{x_B} \sqrt{1 + [y'(x)]^2}\, dx}$$

$f[y, y', x] = 1 + y'(x)$

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2. Now consider a new path Y between point A and point B shown in Fig. 2. The new path is:

$$Y(x) = y(x) + \alpha\eta(x)$$

and when $\alpha = 0$, we get the "correct path" $y(x)$.

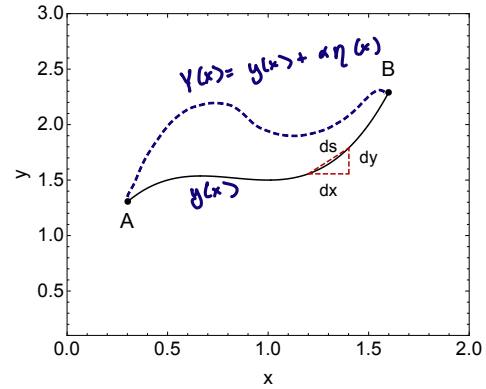

Figure 2: The path between Point A and Point B.

(a) Now L for path Y is in the following form:

$$L = \int_{x_1}^{x_2} f(Y, Y', x)dx$$

Use the chain rule to find:

$$\left.\frac{\partial f(Y, Y', x)}{\partial \alpha}\right|_{\alpha=0}$$

and in terms of $\partial f/\partial y$ and $\partial f/\partial y'$.

$$\frac{\partial f}{\partial \alpha} = \frac{\partial f}{\partial Y}\frac{\partial Y}{\partial \alpha} + \frac{\partial f}{\partial Y'}\frac{\partial Y'}{\partial \alpha} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial \alpha}$$

$$Y = y + \alpha\eta$$
$$Y' = y' + \alpha\eta'$$

$$\Rightarrow \frac{\partial Y}{\partial \alpha} = \eta, \quad \frac{\partial Y'}{\partial \alpha} = \eta', \quad \frac{\partial x}{\partial \alpha} = 0$$

$$\frac{\partial f}{\partial \alpha} = \eta\frac{\partial f}{\partial Y} + \eta'\frac{\partial f}{\partial Y'} + 0 \Big|_{\alpha=0}$$

$$\frac{\partial f}{\partial \alpha}\Big|_{\alpha=0} = \eta\frac{\partial f}{\partial y} + \eta'\frac{\partial f}{\partial y'}$$

b.c $\quad Y|_{\alpha=0} = y, \quad Y'|_{\alpha=0} = y$

(b) In order to minimize L, we want to find the right y such that:

$$\left.\frac{\partial L}{\partial \alpha}\right|_{\alpha=0} = 0$$

Given your answer to part (a), write down $\partial L/\partial \alpha$

$$\frac{\partial L}{\partial \alpha}\Big|_{\alpha=0} = \int_{x_1}^{x_2} \frac{\partial f}{\partial \alpha}\, dx = \boxed{\int_{x_1}^{x_2} \left[\eta\frac{\partial f}{\partial y} + \eta'\frac{\partial f}{\partial y'}\right] dx}$$

(c) Use integration by parts to exchange η' for η, and show that L is minimized when

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0$$

This is the Euler-Lagrange equation.

$$\frac{\partial L}{\partial \alpha} = \int_{x_1}^{x_2} \eta\frac{\partial f}{\partial y} + \underbrace{\int_{x_1}^{x_2} \eta'\frac{\partial f}{\partial y'}\, dx}_{\text{integrate by parts}} = \int_{x_1}^{x_2} \eta\frac{\partial f}{\partial y} + \underbrace{\cancel{\eta\frac{\partial f}{\partial y'}\Big|_{x_1}^{x_2}}}_{\eta(x_2) = \eta(x_1) = 0} - \int \eta\frac{d}{dx}\left[\frac{\partial f}{\partial y'}\right] dx$$

$$\frac{\partial L}{\partial \alpha} = \int_{x_1}^{x_2} \left[\eta\frac{\partial f}{\partial y} - \eta\frac{d}{dx}\frac{\partial f}{\partial y'}\right] dx = \int_{x_1}^{x_2} \eta\left[\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'}\right] dx = 0$$

Note: η is arbitrary, $\eta \neq 0$ in general so $\boxed{\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0}$

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3. Use the Euler-Langrange equation to show that a straight line is the shortest path between point A and B.

4. Find the path that minimizes the time for a rollercoaster to get from point B to point A assuming gravitational acceleration g.

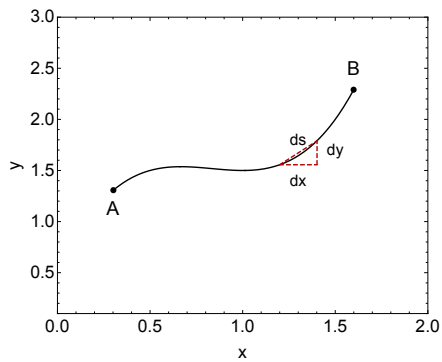

Figure 3: The path between Point A and Point B.

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$$L = \int \sqrt{1 - y'(x)}\, dx$$

$$f = \sqrt{1 - y'(x)}$$

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0$$

$$0 - \frac{d}{dx}\left[\cancel{\tfrac{1}{2}}(1 - [y']^2)^{-1/2}(-\cancel{2}y')\right] = 0$$

$$\int \frac{d}{dx}\left[y'(1 - (y')^2)^{-1/2}\right] dx = \int 0\, dx$$

$$y'(1 - (y')^2)^{-1/2} = c$$

$$y' = c\sqrt{1 - (y')^2}$$

$$(y')^2 = c^2(1 - (y')^2)$$

$$(y')^2 = c^2 - c^2(y')^2$$

$$(y')^2(1 + c^2) = c^2$$

$$y' = \frac{c}{\sqrt{1 + c^2}} \equiv \tilde{c}$$

$$\int y'(x)\, dx = \int \tilde{c}\, dx$$

$$y(x) = \tilde{c}x + b$$

$$\boxed{y = mx + b}$$

(4) $t = \int_B^A \frac{ds}{v}$ ← speed, use conservation of energy

$$\frac{1}{2}mv^2 + mgy = E_T$$

Assume $v = 0$ at point B $\Rightarrow$ $E_T = mgy_B$

$$\frac{1}{2}\cancel{m}v^2 = \cancel{m}g(y_B - y)$$

$$v = \sqrt{2g(y_B - y)}$$

$$t = \int \frac{\sqrt{dx^2 + dy^2}}{\sqrt{2g(y_B - y)}} = \int \underbrace{\frac{\sqrt{x'(y)^2 + 1}}{\sqrt{2g(y_B - y)}}}_{f(x, x', y)} dx$$

$$\frac{\partial f}{\partial x} + \frac{d}{dy}\frac{\partial f}{\partial x'} = 0$$

$$\frac{\partial f}{\partial x} = 0 \qquad \frac{\partial f}{\partial x'} = \frac{\frac{1}{2}(x'(y)^2 + 1)^{-1/2}\, 2x'(y)}{\sqrt{2g(y_B - y)}}$$

$$0 + \frac{d}{dy}\frac{\partial f}{\partial x'} = 0$$

$$\Rightarrow \frac{\partial f}{\partial x'} = \text{constant} \equiv a$$

$$\frac{x'(y)}{\sqrt{2g(y_B - y)(x'(y)^2 + 1)}} = a$$

$$x'(y)^2 = a^2\, 2g(y_B - y)\,[x'(y)^2 + 1]$$

$$x'(y)^2 = a^2\, 2g(y_B - y) + a^2\, 2g(y_B - y)\, x'(y)^2$$

$$x'(y)^2 = \frac{a^2\, 2g(y_B - y)}{1 - a^2\, 2g(y_B - y)}$$

$$\int \frac{dx(y)}{dy}\,dy = \int a \sqrt{\frac{2g(y_B-y)}{1-2a^2g(y_B-y)}}\,dy$$

$$x(y) = \frac{1}{2\sqrt{2}a^2}\sqrt{\frac{1}{g(1-2a^2g(y_B-y))}}$$

$$\times\left[2a(1-2a^2g(y_B-y))\sqrt{y_B-y} - \sqrt{\frac{2}{g}-4a^2(y_B-y)}\,\sin^{-1}\left(\sqrt{2a^2g(y_B-y)}\right)\right] + C$$

$$x(y) = \frac{1}{2\sqrt{2}a^2}\sqrt{\frac{1}{g[1-2a^2g(y_B-y)]}}\left[2a[1-2a^2g(y_B-y)]\sqrt{y_B-y} - \sqrt{\frac{2}{g}[1-2a^2g(y_B-y)]}\,\sin^{-1}\sqrt{2a^2g(y_B-y)}\right] + C$$

$$x(y) = \frac{1}{2\sqrt{2}a^2}\frac{1}{\sqrt{g}}\left[2a\sqrt{1-2a^2g(y_B-y)} - \sqrt{\frac{2}{g}}\,\sin^{-1}\left(2a^2g(y_B-y)\right)\right] + C$$

$$x = \underbrace{\frac{\sqrt{1-2a^2g(y_B-y)}}{a\sqrt{2g}}}_{f(y)} - \frac{1}{2a^2g}\sin^{-1}\left(2a^2g(y_B-y)\right) + C$$

$$2a^2g\,(x-f(y)) = \sin^{-1}\left(2a^2g(y_B-y)\right) + C$$

$$\sin\left(2a^2g(x-f(y))\right) = 2a^2g(y_B-y) + C$$

$$y = 2a^2gy_B + C + \sin\left[2a^2g(x-f(y))\right]$$

$\Rightarrow$ transcendental equation, can analyze parametrically or numerically.

see ex 6.2 of book for more info.